



Cost-Benefit Analysis of Strategic Green Transformation in Line with Sustainable Development Goals (SDGs)



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Abstract: This study aims to test the possible changes in the cost, expense, and income of enterprises that continue their production with traditional methods if they adopt green transformation. Based on sustainable development and green accounting issues that incorporate environmental costs into traditional economic analyses, a comparative cost-benefit analysis was conducted. This analysis examined the revenue impacts, including key cost components such as raw materials, labor, and energy, as well as non-monetary benefits like improved sustainability and social responsibility. The analysis revealed that green transformation could lead to increased upfront costs, but these were offset by long-term benefits such as lower energy expenses, reduced environmental impact, and improved corporate reputation. The study emphasized that green transformation was both a strategically and economically viable option for sustainable growth. The cost-benefit analysis evidenced the strong potential of green transformation in the creation of numerous non-monetary benefits to protect natural resources and to ensure a sustainable future for generations. The findings contribute to the sustainable development literature by highlighting the importance of green transformation in achieving both economic and environmental goals.

Keywords: Green transformation; Sustainability; Cost-benefit analysis; Green accounting; Environmental impact

1. Introduction

Green transformation is a process operating with an objective to ensure environmental sustainability by integrating renewable energy sources, green technologies, and sustainable practices into various sectors such as energy, transport, agriculture, and urban planning. The concept, which was introduced in the 1980s, is frequently administered in every field, especially in the 21st century. It is a participatory process involving the prudent use of the social, cultural, scientific, and natural resources of the society (Bozoğlu Batı, 2024). It aims to reconcile economic development with ecological balance, promote resource efficiency and social equality, and minimize carbon emissions and ecological footprints (Taşel & Bayarçelik, 2023).

Green transformation involves the development and implementation of policies and strategies for transition to a production model. This model could preserve natural capital, apply it efficiently, and prevent pollution, while maintaining economic growth (Durmuşlar et al., 2023). Besides, it involves the development and implementation of legal regulations to mandate energy conservation and reduce greenhouse gas emissions, as well as changing societal attitudes towards more environmentally friendly technological solutions and legal norms (Cheba et al., 2022). It is characterized by both top-down approaches encompassing government and business alliances and bottom-up initiatives driven by grassroots innovative, entrepreneurial, and social organizations (Scoones et al., 2015). In respect of financing, green transformation induces lenders to invest in companies that practise environmental sustainability, to be in support of green transformation policies and controls (Oktasari et al., 2021). The ultimate goal is to guarantee a high-quality life for the present and future generations through the efficient and rational utilization of available resources.

The aim of this study is to predict changes in new cost, expense, and income items in the event that enterprises

continuing their production with traditional methods or classical production forms would shift their emphasis on green transformation. In this way, it is possible to draw the attention of readers or industrialists by analyzing the cost and income items arising from green transformation in the same enterprise, same capacity, and same sector. In cost analysis, monetary as well as non-monetary items are important and both should be taken into consideration.

Green transformation is closely aligned with many of the United Nations Sustainable Development Goals (SDGs), with a particular focus on environmental sustainability, economic growth, and responsible use of resources. The SDGs most relevant to green transformation are set out below:

- *Goal 7: Affordable and Clean Energy:* The green transition boosts the transition from fossil fuels to renewable energy sources, which directly supports this goal by providing access to sustainable and modern energy.
- *Goal 8: Decent Work and Economic Growth:* Cost-benefit analysis of the green transition contributes to promoting sustainable economic growth and productive employment, especially in sectors affected by green technologies.
- *Goal 9: Industry, Innovation, and Infrastructure:* The focus on green transformation and innovation in industry supports the development of sustainable industrialization and resilient infrastructure, which is at the core of this goal.
- *Goal 12: Responsible Consumption and Production:* Green transformation is aligned with this goal by advocating the application of recycled materials, improving sustainable resource management, and reducing environmental impacts through green practices.
- *Goal 13: Climate Action:* Promoting green transformation contributes to climate action by reducing carbon emissions, advocating for renewable energy, and supporting sustainability in production processes.
- *Goal 15: Life on Land:* Green transformation indirectly supports the conservation and sustainable use of terrestrial ecosystems by promoting practices that reduce environmental degradation and loss of biodiversity.

Collectively, these goals emphasize the importance of integrating sustainability into economic and industrial practices, which is at the heart of green transformation.

2. Theoretical Framework

The concept of circular economy, which emphasizes the reuse, recycling, and sustainable management of resources, represents a critical aspect of green transformation. This approach minimizes waste and encourages the efficient use of resources (Aytaç & Yaşar, 2022).

The circular economy approach provides a broader strategic foundation for green transformation by replacing the traditional linear model of “take, make, and dispose” with a regenerative system based on reducing, reusing, and recycling resources. Heshmati (2017) explains that circular economy is designed to address environmental degradation and resource scarcity by promoting closed-loop resource flows, renewable energy use, waste reduction, and the creation of value from reused materials. This framework is directly relevant to green transformation because it links environmental protection with economic efficiency, resource productivity, and long-term sustainable development.

The adoption of renewable energy sources and efficient energy systems is very important in ensuring sustainability. For example, the implementation of solar power plants in the textile industry significantly reduces both production costs and environmental impacts by reducing carbon emissions and dependence on fossil fuels (Koç et al., 2022). Integration of green technologies into production processes helps to reduce environmental impacts. For example, in China, significant progress has been made in reducing emissions and increasing efficiency through a focus on green production (Xu et al., 2021).

The transition to a low-carbon economy also requires a diversified portfolio of mitigation technologies rather than reliance on a single technological solution. Tavoni et al. (2012) show that the feasibility and cost-effectiveness of climate policy depend strongly on the availability and development of low-carbon technologies, particularly renewables and carbon capture and storage. Their findings indicate that technological constraints may increase the cost of climate stabilization, while innovation, learning, and R&D investment can reduce the economic burden of the transition. This insight reinforces the relevance of considering technology-related costs and long-term innovation benefits within the cost-benefit framework of green transformation.

Changing societal attitudes towards sustainability is possible by raising awareness and recommending a culture that values education and environmental stewardship (Cheba et al., 2022). Although there is not yet a thorough and practical national green growth strategy in Türkiye, the concept of green transformation has become widespread and included in policy documents since 2004, in line with efforts to harmonize with the European Union (Durmuşlar et al., 2023). Effective policies and regulations are vital to maintain environmental protection and endorse sustainable practices. In Türkiye, green tax policies significantly regulate environmental sustainability and raise environmental awareness (Dursun & Tutcu, 2024).

Green industrial policies are designed to support the transformation of traditional industries into sustainable

industries. It includes promoting the development of eco-industrial parks that aid the clustering of industries to minimize environmental impact through shared resources and waste management systems (Zeng et al., 2021).

Green industrial policy literature also emphasizes that the benefits of green transformation go beyond environmental protection. Altenburg & Assmann (2017) argue that green industrial policies may generate important co-benefits for developing economies, including resource efficiency, energy savings, improved health conditions, employment creation, stronger energy security, and new competitive advantages in environmental goods and technologies. This perspective supports the argument that green transformation should be evaluated not only as a cost-increasing adjustment but also as a strategic opportunity that may improve long-term productivity, competitiveness, and sustainability.

Recent evidence also indicates that the financing structure of green investments is a critical determinant of their economic and environmental outcomes. Křístková et al. (2025) compare European Green Bonds, carbon-tax financing, and crowding-out scenarios and find that green investments generally generate positive GDP, social, and emission-saving effects when they do not displace other investments. Their findings also show that green bonds can support the transition by mobilizing additional investment resources, while carbon taxes may accelerate emission reductions but require careful policy design to avoid adverse social and economic effects. This evidence reinforces the present study's argument that tax incentives, green finance mechanisms, and coordinated public support are essential for making green transformation economically feasible for enterprises.

The Green Deal Action Plan published by the Ministry of Trade in 2021 sets out clear green transformation targets for Türkiye, particularly in line with the transformation policies of the European Union and other countries. Under the slogan of "A Green and Circular Economy", the plan includes various actions to be prepared for the impacts that may occur during the process of implementing Green Deal and to strengthen the green transformation infrastructure (Republic of Türkiye Ministry of Trade, 2021).

It will take a long time for the investments made for green transformation to provide economic, social, and environmental gains and to observe the effects of improvement in natural resources. In this context, recognizing the short-term effects of transformation is of great importance in order to motivate the relevant stakeholders for long-term goals and to ensure that they adopt the transformation (İzmir Development Agency, 2022).

The first international regulation accepted in the literature on environmental policies is the Kyoto Protocol, which was adopted in December 1997 with the objective of combating climate change. It established an international framework for the countries that accepted the protocol to reduce greenhouse gas emissions and limit the effects of global warming (United Nations Framework Convention on Climate Change). Another international regulation is the Paris Agreement, which was organized in December 2015 and entered into force in 2016. This agreement aims to direct all nations to combat climate change (Dursun & Tutcu, 2024).

The European Green Deal, published by the European Commission in December 2019, aims to turn Europe into a region with net zero carbon emissions by 2050. This Consensus, which also includes Türkiye, is not only a climate action plan but also an all-inclusive economic plan that includes issues such as clean and secure energy supply, circular economy, smart cities, green financing, carbon tax, and job creation (Aydınoğlu & Özdemir, 2022). Within the scope of the European Climate Conference, in line with the objectives of the United Nations Sustainable Development Agenda and the Paris Climate Agreement, it is targeted to "transition to a competitive and green economy within the limits of the planet". As part of this transition, policies have been developed for sustainable consumption and for consumers in the European Union to take a more active role in green transformation (Veral, 2023).

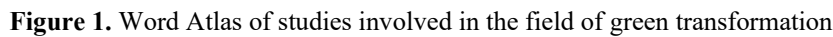
Koç et al. (2022) stated that the textile sector, which had an important share in exports and total energy consumption, should switch to environmentally friendly systems by reducing production costs in order to survive in a rapidly developing competitive environment and to adapt to changing export conditions within the framework of the "Green Deal". The application of solar energy, one of the renewable energy sources, in a yarn production factory was investigated and the effects of this application on reducing costs in factory production and its concomitant environmental benefits were identified.

As a result of global developments and international agreements, sustainable development has become a necessity for businesses beyond being an initiative. Companies around the world have started to implement green transformation as a result of legal regulations (Bozoğlu Bati, 2024).

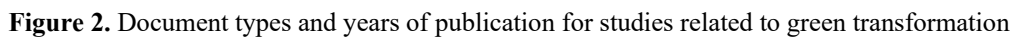
Green accounting practices cover issues such as directing investments by taking into account environmental impacts, measuring and analyzing environmental costs, identifying potential risks, re-evaluating assets, protecting capital, preparing environmental improvement programs, and making cost analyses. It also includes studies for the development of ecological accounting systems (Bağdat, 2024):

- Encouraging public institutions and businesses to take effective environmental actions to achieve sustainable development;
- Embedding the behavior of efficient consumption of natural resources into the economic system of the country;
- Systematizing environmental improvement works and ensuring cost minimization;
- Ensuring the integration of environmental studies into financial statements;

- The literature review of the study was carried out by accessing through lens.org, Google Scholar and Trakya University online library. In the search box, “green transformation” was typed in English to search within the titles only. According to the findings, 829 studies and 9,312 citations in total were obtained. The Word Atlas created in accordance with the literature review is given in Figure 1.



The types of studies and their publication status by year are given in Figure 2.



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in journal articles.

The following findings were obtained in the scanning according to country groups:

- 102 studies were conducted in Organisation for Economic Co-operation and Development (OECD) countries and 2,870 citations were received in total.
- In BRICS countries, 273 studies were conducted and 5,485 citations were received in total.
- 50 studies were conducted in European Union countries and 1,812 citations were received in total.
- 336 studies were conducted in G20 countries and 6,698 citations were received in total.
- 58 studies were conducted in G7 countries and 2,152 citations were received in total.

A general view of countries working in the field of green transformation in the world is given in Figure 3.

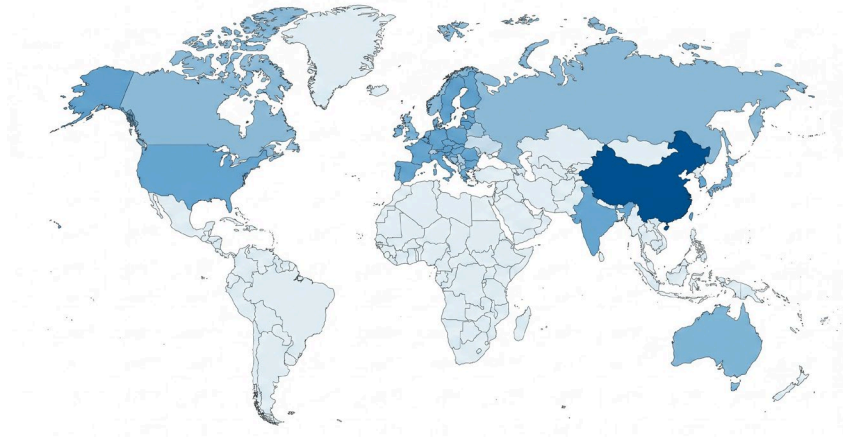


Figure 3. Global distribution of studies in the field of green transformation

In the figure, dark areas suggest that more studies have been conducted, while light areas illustrate the occurrence of fewer studies. When the figure was interpreted, the country with the highest number of studies (247 studies) was China. It was followed by 23 studies conducted in the United Kingdom and 16 studies in Germany. Türkiye had only conducted 3 studies, yet the studies conducted in this country received 76 citations.

In the examination of the Google Scholar database, the word group “Green transformation” was searched in the titles. According to the findings, 1,290 studies were found.

The most cited study is Altenburg and Pegels’ article titled “Sustainability-oriented innovation systems—Managing the green transformation” published in *Innovation and Development* in 2017 with 230 citations (Altenburg & Pegels, 2017). The paper introduced the concept of Sustainability Oriented Innovation Systems (SoIS) as a framework for addressing environmental challenges such as global warming. SoIS emphasizes the need for governance to disrupt unsustainable technological pathways and elevate alternative technologies. The two scholars emphasized the importance of a green techno-economic shift that included technological innovation as well as changes in social norms and economic incentives to internalize environmental costs. The authors also argued that national policies significantly influenced technological trajectories and led to different pathways in sustainable technology development. The study extrapolated that radical policy changes and international co-operation were required to facilitate a sustainable technological transformation.

The second most cited study with 117 citations is “Natural resource consumption and industrial green transformation: Does the digital economy matter?” by Ran et al. (2023) published in *Resources Policy*. The study investigated the role of digital economy in promoting industrial green transformation by analyzing data from 30 Chinese provinces between 2006 and 2020. The study revealed that digital economy considerably enhanced industrial green transformation by increasing the consumption of natural resources and the marked regional differences in the impacts. In the east-central regions of China, the digital economy effectively drove green transformation through improved utilization of resources, while this effect was less significant in the western regions. The findings manifested that as the digital economy grew, its positive impact on green transformation strengthened. Policymakers are advised to adapt digital economy policies to improve the utilization efficiency of natural resources and support green transformation in industry, taking into account regional differences.

The third most cited study with 75 citations is the book chapter “Green transformation” by Schmitz (2015) in the book *The Politics of Green Transformations*. The study explored the urgent need for green transformation to address climate change. It argued for redefining the discourse from “climate change” to “climate chaos” to better match the public’s experience of extreme weather. The importance of strategic alliances between governments, businesses, and civil society was emphasized to initiate transformation, with interests going beyond environmental sustainability to include motives, such as energy security and economic development.

3. Methodology

The evaluation procedure integrated monetary and non-monetary factors by first identifying standard traditional production costs and juxtaposing them with their green transformation equivalents. Non-monetary items (e.g., environmental destruction, and livability) were qualitatively grouped, based on their presence or absence in traditional versus green paradigms. These grouped factors were then compiled into a comparative matrix to illustrate the overarching strategic benefits alongside the estimated monetary shifts, without relying on advanced quantitative modeling.

An integrated evaluation approach is particularly important in green economy and green transformation studies because policy and investment decisions may generate simultaneous economic, social, and environmental effects. Bassi (2015) argues that the Green Economy Model was developed to overcome sectoral policy silos by linking physical, human, social, and natural capital through dynamic feedbacks. This perspective supports the methodological structure of the present study, which considers both monetary and non-monetary factors when comparing traditional production methods with green transformation practices.

This study employed a comparative cost-benefit analysis to assess the financial and environmental impacts of adopting green transformation practices in enterprises.

The methodology was structured as follows:

A comprehensive review of the existing literature on green transformation and sustainability with cost-benefit analysis was conducted. The literature review was drawn from databases such as Google Scholar, Lens.org and Trakya University online library, with a focus on studies published in peer-reviewed journals, books, and conference proceedings. This review enabled an understanding of the key concepts and the identification of the main cost and revenue components related to green transformation.

The details of the cost and expenditure components that may be related to green transformation were examined and analyzed, according to the account codes and names in the uniform chart of accounts required to be used by the Ministry of Finance of Türkiye. The accounts' codes start from 710 and continue until 780. Drawing on the literature, the study identified the main cost and revenue items related to green transformation. These included direct costs such as raw materials, labor, and energy as well as indirect costs such as operational disruptions during the transition period. Revenue components include direct financial gains from energy savings and reduced material costs, as well as non-monetary benefits such as improved corporate reputation and compliance with environmental regulations.

Through cost-benefit analysis, the identified costs and benefits of green transformation were tabulated and compared with traditional production methods. Where possible, monetary values were assigned to environmental benefits for both options to determine the long-term financial viability of green transformation. Revenue and cost items in the study included both monetary and non-monetary factors.

In addition, cost-benefit analysis in green transition studies can be strengthened by expressing environmental benefits in monetary terms. Alpino et al. (2023) evaluate green investments in the Italian National Recovery and Resilience Plan by comparing investment costs with discounted environmental benefits calculated through the social cost of carbon. Their approach shows that the economic desirability of green investments depends not only on direct costs but also on the timing of expected emission reductions, the selected discount rate, and the monetary valuation of avoided environmental damages. This perspective supports the present study's emphasis on evaluating green transformation through both monetary and non-monetary benefits.

The analysis revealed the returns and potential long-term benefits of adopting sustainable practices. This methodological approach provided a framework for assessing the financial and environmental impacts of green transformation and highlighted its applicability as a strategy for achieving sustainability in business.

To determine monetary values for the cost-benefit analysis, the numerical values used in the comparative framework represent an illustrative scenario rather than empirical data derived from a specific real-world case study. These figures were intended solely to demonstrate the analytical framework of the comparative analysis and to illustrate the relative shifts in cost components (e.g., raw materials, labor, and energy) between traditional and green production methods. It is important to note that these values do not represent actual industrial conditions but serve as a theoretical baseline to help readers understand the overarching cost-benefit structure.

4. Results and Discussion

The findings outlined in this section were primarily derived from the interpretation of literature and conceptual assumptions regarding traditional versus green production methods, rather than direct empirical evidence. The following evaluation explicitly distinguished between theoretically anticipated outcomes and the comparative framework designed to assess them.

In this study, the concept of green transformation was also defined in terms of accounting. In comparison with traditional production, the kind of costs incurred by green transformation for businesses and the advantages it could provide have been highlighted. Before proceeding to the cost-benefit analysis, the proposed cost and income

elements of green transformation practices were emphasized and itemized below:

Cost and income elements of green transformation practices:

- 710 Direct raw materials and supplies expenses
 - Using high technological raw materials: Although it increases the cost, it can contribute to the production of smart products. In addition, it can enable production that will contribute to sustainability such as energy saving, self-cleaning, and the like.
 - Using recycled raw materials: Focusing on the use of recycled materials in production will enhance sustainability.
 - Using abundant substitutes instead of scarce raw materials: Reducing dependence on scarce resources will increase sustainability.
- 720 Direct labor expenses
 - Implementing labor policies attaching importance to workers' health: Prioritizing workers' health will have a positive impact on health.
 - Implementing labor policies attaching importance to occupational safety: Improving safety measures in the workplace will have a positive impact on health.
 - Using robots instead of labor (e.g., in the denim bleaching process): Reducing workers' exposure to harmful processes will lead to improved health.
 - Using workers instead of robots sectorally: Employing humans rather than robots in every possible field will increase business opportunities by affecting employment.
 - Implementing processes that reduce time losses: Designing and developing business processes to reduce time losses will increase productivity by reducing working time.
- 730 General production expenses
 - Consumption of renewable energy: Green transformation practices will contribute to sustainability by reducing the carbon footprint and saving on carbon footprint payments.
 - Not using fossil fuels: It will increase sustainability by reducing emissions.
 - Basing processes on muscle power: It will increase employment opportunities.
- 750 Research and development expenses
 - Conducting R&D on green transformation: It will promote sustainability by investing in environmentally friendly innovations.
- 760 Marketing, sales, and distribution expenses
 - Reducing stationery-based processes: It will support sustainability by reducing the consumption of paper and protecting forest assets.
 - Utilizing renewable energy: Advocating sustainability in marketing and distribution operations.
 - Not using fossil fuels: Promoting sustainability by reducing dependence on fossil fuels.
 - Basing processes on muscle power: Increasing business opportunities by impacting the employment.
- 770 General administrative expenses
 - Reducing stationery-based processes: It will increase sustainability by reducing the consumption of paper.
 - Consuming renewable energy: It will support sustainability in management function.
 - Not employing fossil fuels: It will increase sustainability by reducing fossil fuel consumption.
 - Basing processes on muscle power: It will positively impact employment by increasing business opportunities.
- 780 Financing expenses
 - Using loans from institutions that give importance to the green transformation process: It will support sustainability by encouraging financial resources compatible with environmentally friendly investments.

Recent reviews of low-carbon and sustainable technologies further support the relevance of energy saving, emission reduction, renewable energy applications, district heating systems, and economic assessment in green transformation. Chu et al. (2022) emphasize that sustainable energy strategies should combine efficiency improvements, renewable energy integration, advanced energy storage, and supportive policies to reduce emissions and improve economic feasibility. This perspective is consistent with the present study's comparative framework, as it shows that green transformation practices should be evaluated not only through direct cost changes but also through energy efficiency, emission reduction, and long-term sustainability benefits.

The cost-benefit analysis of green transformation is given in Table 1 and Table 2.

Mapping green transformation practices to SDGs: The identified green transformation practices were directly mapped onto the SDGs mentioned earlier. For instance, transitioning from fossil fuels to renewable energy (Account 730) operationalized Goal 7 (Affordable and Clean Energy) and Goal 13 (Climate Action). The use of recycled raw materials (Account 710) directly supported Goal 12 (Responsible Consumption and Production). Furthermore, sustainable labor policies (Account 720) aligned with Goal 8 (Decent Work and Economic Growth).

This alignment demonstrated how strategic accounting changes inherently supported global sustainability targets.

Table 1 presents in detail the impacts of various business processes and decisions related to green transformation on sustainability, health, employment, and productivity. The table also exposes the health and employment impacts of sustainable practices such as using renewable energy and recycled materials.

Table 1. Green transformation practices based on cost and income items and their impact on accounting

Cost Accounts	Implementation of Green Transformation	Impact
710 Direct raw materials and supplies expenses	Using high technological raw materials	Sustainability
	Using recycled raw materials	Sustainability
	Using abundant substitutes instead of scarce raw materials	Sustainability
	Implementing labor policies attaching importance to workers' health	Health
720 Direct labor expense	Implementing labor policies attaching importance to occupational safety	Health
	Using robots instead of labor (e.g., in the denim bleaching process)	Health
	Using workers instead of robots sectorally	Employment
	Implementing processes that reduce time losses	Reduces working time
730 General production expenses	Use of renewable energy	Sustainability
	Not using fossil fuels	Sustainability
	Basing processes on muscle power	Employment
750 Research and development expenses	Conducting R&D on green transformation	Sustainability
760 Marketing, sales, and distribution expenses	Reducing stationery-based processes	Sustainability
	Use of renewable energy	Sustainability
	Not using fossil fuels	Sustainability
	Basing processes on muscle power	Employment
	Reducing stationery-based processes	Sustainability
770 General administrative expenses	Use of renewable energy	Sustainability
	Not using fossil fuels	Sustainability
	Basing processes on muscle power	Employment
780 Financial expenses	Using loans from institutions that give importance to the green transformation process	Sustainability

Cost and revenue aspects of green transformation and traditional production:

1. Revenues

a. Non-monetary revenues

(i) A livable world

Traditional production: This method of production does not ensure a habitable world. Excessive consumption of natural resources and environmental damage will reduce livability.

Green transformation: It will contribute to a livable world by preserving environmental sustainability and ecological balance.

(ii) Future concern of living things

Traditional production: Traditional production methods will pose environmental risks and concerns of resource scarcity for future generations.

2. Costs

a. Monetary costs

(i) Natural resource consuming raw materials

Traditional production: At a cost of 1,000 units, raw materials that consume natural resources are used.

Green transformation: The use of recycled or substitute raw materials will reduce costs to 500 units.

(ii) Labor policies

Traditional production: Costs are 500 units due to unsustainable labor policies.

Green transformation: Implementing sustainable labor policies may increase costs to 1,000 units, but this will benefit workers' health and productivity in the long run.

(iii) Energy consumption

Traditional production: Fossil fuel consumption costs 1,000 units.

Green transformation: Using renewable energy will reduce costs to 500 units and reduce environmental impacts.

b. Non-monetary costs

(i) Destroyed nature

Traditional production: There will be costs such as environmental destruction and destruction of natural areas.

Green transformation: Natural areas will be protected and environmental destruction will be prevented.

(ii) Vanishing new generation

Traditional production: It will pose a risk of environmental and resource loss for future generations.
Green transformation: The quality of life of future generations will be protected.

3. Total costs

Traditional production: Total costs are 3,000 units.

Green transformation: Total costs are 3,000 units.

This result reflects the existence of a more sustainable production process at the same cost.

Table 2 unveils that green transformation is more advantageous in both environmental and economic terms. While traditional production methods lead to the same cost amount and environmental damage, green transformation will be able to usher in a sustainable future with the same amount of cost in the long run. Green transformation will also allow businesses to fulfill their social responsibilities.

Table 2. Cost-benefit analysis of green transformation

Category	Type	Item	Traditional Production	Green Transformation*
Revenues	Non-monetary income	Use of high-tech raw materials to protect the environment	No	Yes
		A livable world	No	Yes
		The future concern of living things	Yes	No
		Use of high technological raw materials	-	1000
		Use of traditional raw materials	500	-
Costs	Monetary costs	Raw material consuming natural resources	1000	-
		Use of recycled raw materials/substitute raw materials	-	500
		Unsustainable labor policy	500	-
		Sustainable labor policy	-	1000
	Non-monetary costs	Fossil fuel consumption	1000	-
		Renewable energy consumption	-	500
		Destroyed nature	Yes	No
		Vanishing new generation	Yes	No
Total costs	-	-	3000	3000

Note: The figures given for monetary costs and revenues should be perceived as currency. They can be considered as each country's own currency or universal currencies. The hyphen (-) denotes not applicable.

5. Conclusions

In the literature, concepts such as “environmental accounting”, “green accounting”, “natural resource accounting”, “sustainability accounting” were adopted interchangeably. Green transformation, often referred to as green transition, refers to the restructuring of economic activities to integrate environmentally sustainable practices. This careful approach encompasses technological, economic, and social changes that aim to guarantee long-term environmental sustainability, while promoting economic growth.

With developments in the industry, it is essential to identify and measure the environmental impacts of business activities and to work towards eliminating these risks. With the climate change all over the world and the associated negative effects of these changes, the importance of the concept of green transformation is becoming more widely recognized. Green transformation is the basis for sustainable development.

Switching from production processes based on the traditional system to green transformation production processes does not necessarily mean that production costs will be reduced. Implementing production processes based on green transformation may also result in higher costs for businesses. This is a possible reality, but the opposite can also happen. That is, implementing the new green transformation can reduce costs. Although the increase in costs of green transformation may seem negative for the business, it may create many non-monetary benefits for a sustainable future in our universe where eight billion people live, for the continuation of the human generation and for the protection of natural resources. We characterize it as a non-monetary benefit, it is a much more valuable reality than the most important monetary values. However, it is vital to acknowledge that the actual outcomes and financial impacts of green transformation may vary significantly, depending on specific industry characteristics, available investment conditions, and the scale of implementation. While the long-term benefits are theoretically robust, practical success requires careful strategic planning.

Ensuring that green technologies and practices are economically viable is one of the biggest challenges. It requires supportive policies and financial mechanisms to make sustainable options (such as new materials that reduce environmental footprints) to be competitive with conventional options. For businesses to adopt green transformation practices, providing various tax incentives and financial support to businesses with such practices will render the process preferable. It will also encourage investments in renewable energy, energy efficiency, and other sustainable initiatives. In this light, the realization of green transformation will be possible through “global

cooperation”.

Author Contributions

Conceptualization, S.Y.; formal analysis, S.Y.; writing—original draft preparation, A.A.Ç.; writing—review and editing, S.Y. and A.A.Ç.; supervision, S.Y. All authors have read and agreed to the published version of the manuscript.

Data Availability

Not applicable.

Conflicts of Interest

The authors declare no conflicts of interest.

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